

# Synthesis Report

Family: Cyclone IV

Device: GX EP4CGX150DF31I7AD

Optimization techniques used: Balanced (Normal Flow)

|         | Area            | Dynamic Power | Max Frequency | LUTs   | FFs   | Memory | Setup Slack |
|---------|-----------------|---------------|---------------|--------|-------|--------|-------------|
| Vanilla | 12,897 LEs (9%) | 25.69 mW      | 132.21 MHz    | 12,897 | 8,514 | 0 bit  | 1.038       |

Flow Summary Screenshot (See corelet-Flow Summary.rpt)

The screenshot displays the Quartus Prime IDE interface. The 'Project Navigator' on the left shows the project hierarchy for 'Cyclone IV GX: EP4CGX150DF31I7AD'. The 'Tasks' pane shows the compilation process, including 'Compile Design', 'Analysis & Synthesis', 'Fitter (Place & Route)', 'Assembler (Generate program)', 'Timing Analysis', and 'EDA Netlist Writer'. The 'Table of Contents' pane lists various reports, with 'Flow Summary' selected. The 'Flow Summary' report is displayed in the main window, showing the following details:

- Flow Status: Successful - Sat Dec 13 00:36:52 2025
- Quartus Prime Version: 25.1std.0 Build 1129 10/21/2025 SC Lite Edition
- Revision Name: corelet
- Top-level Entity Name: corelet
- Family: Cyclone IV GX
- Device: EP4CGX150DF31I7AD
- Timing Models: Final
- Total logic elements: 12,897 / 149,760 ( 9 % )
- Total registers: 8514
- Total pins: 427 / 508 ( 84 % )
- Total virtual pins: 0
- Total memory bits: 0 / 6,635,520 ( 0 % )
- Embedded Multiplier 9-bit elements: 0 / 720 ( 0 % )
- Total GXB Receiver Channel PCS: 0 / 8 ( 0 % )
- Total GXB Receiver Channel PMA: 0 / 8 ( 0 % )
- Total GXB Transmitter Channel PCS: 0 / 8 ( 0 % )
- Total GXB Transmitter Channel PMA: 0 / 8 ( 0 % )
- Total PLLs: 0 / 8 ( 0 % )

The bottom pane shows the 'Messages' window with the following messages:

- 223000 Starting Vectorless Power Activity Estimation
- 223001 Completed Vectorless Power Activity Estimation
- 218000 Using Advanced I/O Power to simulate I/O buffers with the specified board trace model
- 334003 Started post-fitting delay annotation
- 334004 Delay annotation completed successfully
- 215049 Average toggle rate for this design is 0.844 millions of transitions / sec
- 215031 Total thermal power estimate for the design is 320.36 mW
- Quartus Prime Power Analyzer was successful. 0 errors, 4 warnings

Max Frequency Screenshot (See corelet-Fmax Summary.rpt)

**Slow 1200mV 100C Model Fmax Summary**

|   | Fmax       | Restricted Fmax | Clock Name | Note |
|---|------------|-----------------|------------|------|
| 1 | 132.21 MHz | 132.21 MHz      | clk        |      |

This panel reports FMAX for every clock in the design, regardless of the user-specified paths where the source and destination registers or ports are driven by the same clock. Generated clocks, are ignored. For paths between a clock and its inversion, FMAX is scaled along with FMAX, such that the duty cycle (in terms of a percentage) is maintained. Always use clock constraints and other clock reports for sign-off analysis.

**Message Log:**

- 223000 Starting Vectorless Power Activity Estimation
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- 215049 Average toggle rate for this design is 0.844 millions of transitions / sec
- 215031 Total thermal power estimate for the design is 320.36 mW
- Quartus Prime Power Analyzer was successful. 0 errors, 4 warnings

Power Analyzer Screenshot (See corelet-Power-Summary.rpt)

**Power Analyzer Summary**

| Power Analyzer Status                  | Successful - Sat Dec 13 00:36:52 2025            |
|----------------------------------------|--------------------------------------------------|
| Quartus Prime Version                  | 25.1std.0 Build 1129 10/21/2025 SC Lite Edition  |
| Revision Name                          | corelet                                          |
| Top-level Entity Name                  | corelet                                          |
| Family                                 | Cyclone IV GX                                    |
| Device                                 | EP4CGX150DF31I7AD                                |
| Power Models                           | Final                                            |
| Total Thermal Power Dissipation        | 320.36 mW                                        |
| Core Dynamic Thermal Power Dissipation | 25.69 mW                                         |
| Core Static Thermal Power Dissipation  | 119.41 mW                                        |
| I/O Thermal Power Dissipation          | 175.27 mW                                        |
| Power Estimation Confidence            | Low: user provided insufficient toggle rate data |

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Timing Closure and Critical Path (See corelet-Slow 1200mV 100C Model .rpt)

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Slow 1200mV 100C Model report for corelet
Sat Dec 13 16:45:07 2025
Quartus Prime Version 25.1std.0 Build 1129 10/21/2025 SC Lite Edition

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; Table of Contents ;
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1. Legal Notice
2. Command Info
3. Summary of Paths
4. Path #1: Setup slack is 1.038
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: Command Info :
Report Timing: Found 1 setup paths (0 violated). Worst case slack is 1.058

 Tcl Command:
report_timing -setup -multi_corner -panel_name {Report Timing} -apaths 1 -detail full_path

Options:
  -setup
  -apaths 1
  -detail full_path
  -panel_name {Report Timing}
  -multi_corner

Delay Model:
Slow 1200nm/ 100C Model

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Summary of Paths
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: Path ID : Setup Mode : Start Node : To Node : Launch Clock : Launch Clock : Relationship : Clock Skew : Data Delay :
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: 1.058 : mac_array_mac_array_inst[mac_row_row_mux[0] mac_row_instance[mac_vile_col_mux[1] mac_vile_instance[h_q[1]] : off0:off0_inst[fifo_deptr36_col_mux[0] fifo_instance[q16[1]] : clk : clk : 10.000 : -0.041 : 8.919 :
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[illegible]